K-Nearest Neighbours

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CM22009 – Machine Learning

This lecture adapts material originally prepared by Dr. Andy Barnes (with permission)

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Recap: Logistic Regression

- Hypothesis function: $h_\theta(x) = g(\theta^T x), g(z)=\frac{1}{1+e^{-z}}$
- Log loss: $J(\theta) = -\frac{1}{m}\sum_{i=1}^{m}\left[y^i \log\left(h_\theta(x^i)\right) + (1-y^i)\log\left(1-h_\theta(x^i)\right)\right]$
- Gradient Descent: repeat this updating for each parameter:

$$\theta_j = \theta_j - \alpha \frac{\partial}{\partial \theta_j} J(\theta) = \theta_j - \alpha \frac{1}{m}\sum_{i=1}^{m}(h_\theta(x^i) - y^i)x_j^i$$

Stop when it reaches a certain cost threshold or number of iterations.

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Where are we?

Week No.	Lecture A	Lecture B
1 (29/09/2025)	Intro. Supervised Learning	Logistic Regression
2 (06/10/2025)	Nearest-Neighbours	Support Vector Machines
3 (13/10/2025)	Decision Trees	Random Forests
4 (20/10/2025)	Naïve-Bayes	Data Engineering
5 (27/10/2025)		
6 (03/11/2025)	Reading Week	

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Learning Objectives

Explain how the k-NN operates for regression and classification tasks.

Describe the advantages and disadvantages of k-NN.

Calculate continuous and categorical distances between two vectors.

Distinguish parametric vs non-parametric, overfitting vs underfitting

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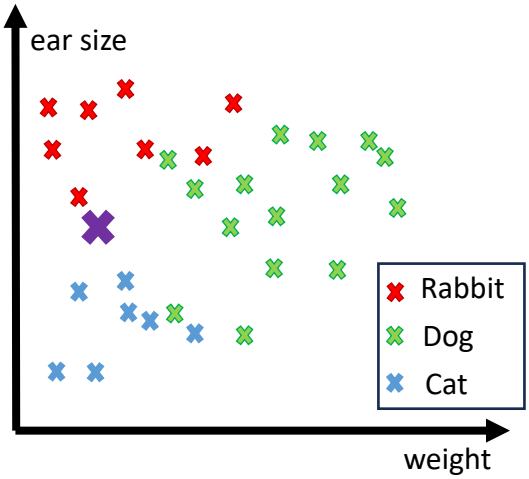
K-Nearest Neighbours

- Often referred to as KNN or k-NN.
- **Intuition**: similar data points are often located near each other in the feature space
- **Approach**: Given a test sample, finding K training samples that are closest to the test sample, then make the prediction based on
  - Majority class (classification)
  - Their average target value (regression)



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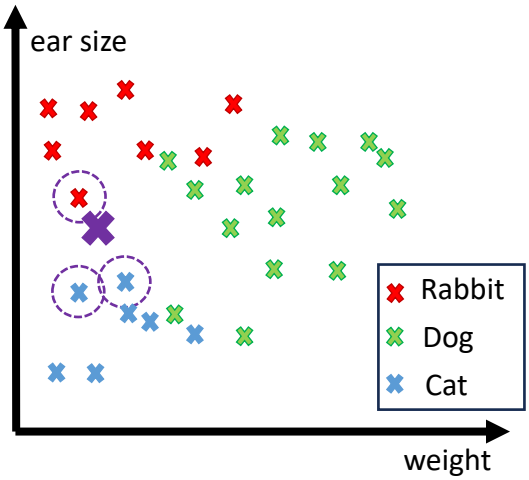
K-Nearest Neighbours

When $K = 1$ ➡ Rabbit

When $K = 3$ ➡ Cat

When $K = 100$ ➡ Dog (the majority class)



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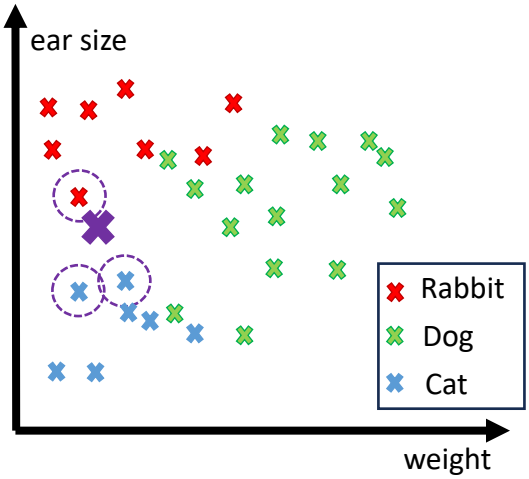
How to choose the right K?

K too small → Sensitive to noise

K too big → Underfitting
(the majority class)



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How to choose the right K?

K too small → Sensitive to noise

K too big → Underfitting
(the majority class)

Cross validation (Best Practice)

K-fold Cross validation

1. Split data into k equal-sized folds (typically 5 or 10)
2. Iterate k times:
   - Use 1 fold as validation set, the remaining $k-1$ folds as training set
   - Record the validation performance on different K value ($K = 3, 5, 7, \ldots$)
3. Average the performance across all k iterations.
4. Choose K with the best average validation performance

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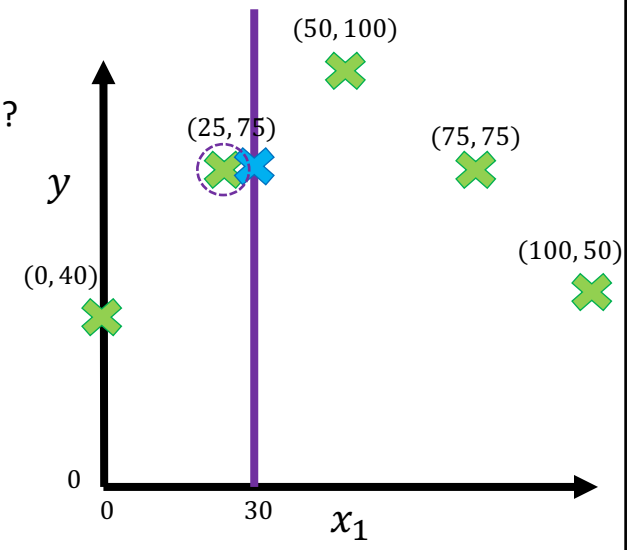
KNN Regression

- How can KNN work with regression?
  - Let's take a 1D feature space and an output (y), we'll set $K = 1$.
  - Let's put a new point at:

$$x_1 = 30 \quad \Rightarrow \quad y = 75$$



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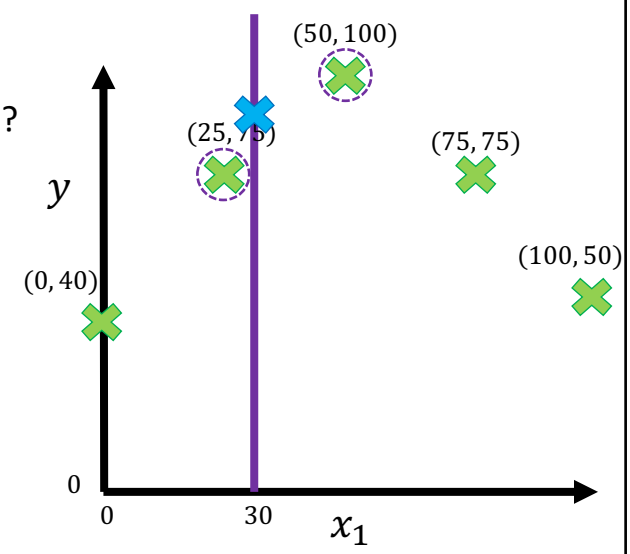
KNN Regression

- How can KNN work with regression?
  - Let's take a 1D feature space and an output (y), we'll set $K = 2$.
  - Let's put a new point at:

$$x_1 = 30 \quad \Rightarrow \quad y = \frac{75 + 100}{2} = 87.5$$



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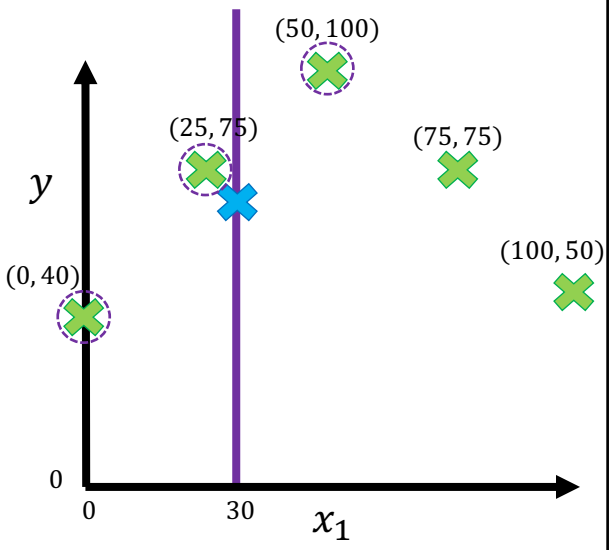
KNN Regression

- How can KNN work with regression?
  - Let's take a 1D feature space and an output (y), we'll set $K = 3$.
  - Let's put a new point at:

$$x_1 = 30 \Rightarrow y = \frac{40 + 75 + 100}{3}$$

$$= 71.6$$

- How about $K = m$ where m is the number of training items?
  - Any prediction will just be the mean of the training set.



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Advantages	Disadvantages
Intuitive and easy to implement	Computationally expensive at testing stage
No 'training' needed (Lazy learning)	Must hold training data in memory
Adaptable (multi-class, non-linear)	Curse of Dimensionality (Most points become equidistance in high dimensions, e.g., 100+ dim)

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When to use KNN?

- Small to Medium datasets
- Low to moderate dimensionality (<20-50 dim)
- Non-linear decision boundaries, relationships between features and target is complex
- Frequent model updates with new data while need immediate deployment

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KNN classification examples

- Recommendation system
- Customer segmentation with distinct groups
- Medical diagnosis: find similar patient cases
- Legal case analysis

KNN regression examples

- House price prediction (based on similar houses)
- Stock price forecasting: find similar historical patterns
- Credit scoring: find similar credit profiles
- Data pre-processing: Missing data interpolation

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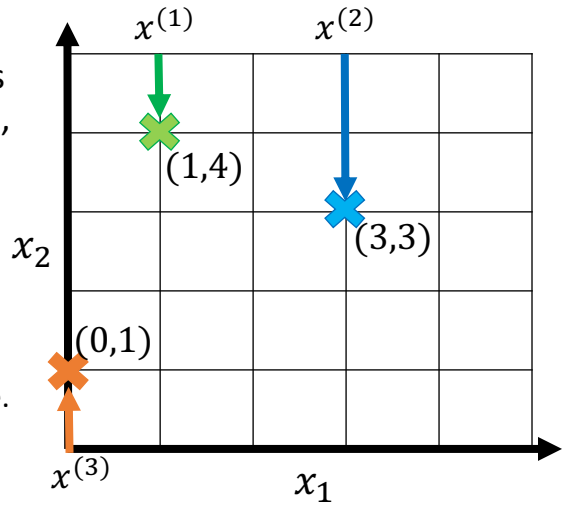
Distance Metrics

- In Machine learning, many algorithms rely on comparing vectors – e.g., KNN, clustering, similarity search, etc.

Q: But what does it mean for two points/vectors to be '**close**' to each other? Can we say $d(x^{(1)}, x^{(2)}) < d(x^{(1)}, x^{(3)})$?

A: This depends on the distance metric we choose.



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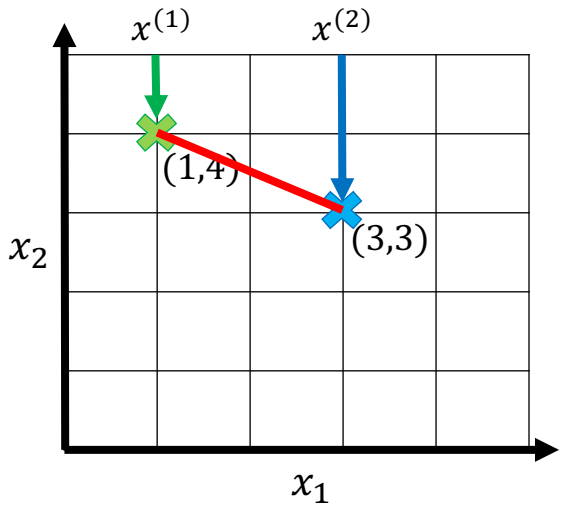
Distance Metrics

Euclidean Distance

Measures the straight-line distance between two points in space



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Distance Metrics

Euclidean Distance

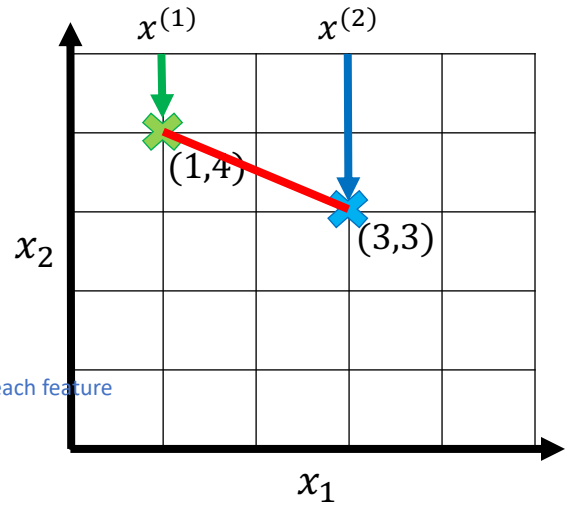
Measures the straight-line distance between two points in space

L_2 norm of their difference

$$\| x^{(1)} - x^{(2)} \|_2 = \sqrt{\sum_{j=1}^{D} (x_j^{(1)} - x_j^{(2)})^2}$$

Take the squared difference on each feature

Square root the total difference



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Distance Metrics

Euclidean Distance

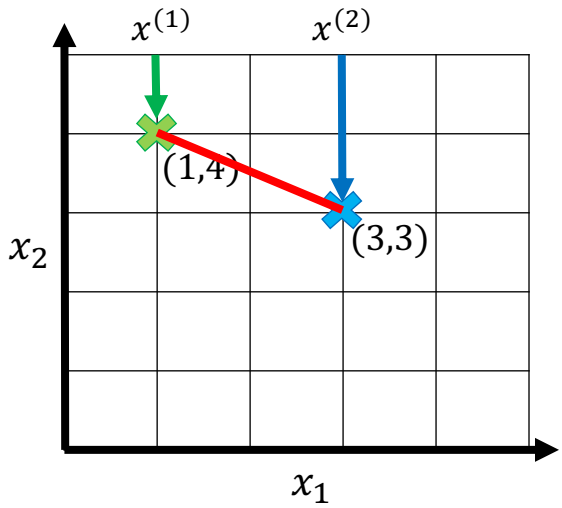
$$\sqrt{\sum_{j=1}^{D} (x_j^{(1)} - x_j^{(2)})^2}$$

$$\sqrt{(x_1^{(1)} - x_1^{(2)})^2 + (x_2^{(1)} - x_2^{(2)})^2}$$

$$\sqrt{(1-3)^2 + (4-3)^2}$$

$$\sqrt{4+1}$$

$$\sqrt{5}$$



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Distance Metrics

Euclidean Distance $\sqrt{5}$

Manhattan Distance

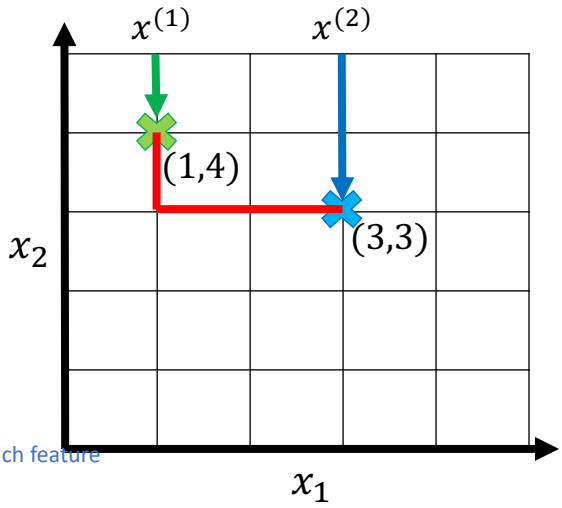
Measures distance as the sum of absolute differences along each dimension

L_1 norm of their difference

$$\| x^{(1)} - x^{(2)} \|_1 = \sum_{j=1}^{D} \left| x_j^{(1)} - x_j^{(2)} \right|$$

Take the absolute difference on each feature

Sum across all dimensions



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Distance Metrics

Euclidean Distance $\sqrt{5}$

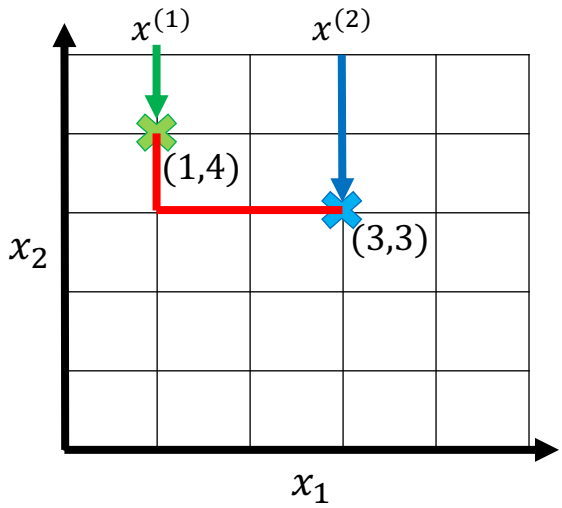
Manhattan Distance

$$\sum_{j=1}^{D} \left| x_j^{(1)} - x_j^{(2)} \right|$$

$$\left| x_1^{(1)} - x_1^{(1)} \right| + \left| x_2^{(1)} - x_2^{(2)} \right|$$

$$2 + 1$$

$$3$$



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Distance Metrics

Euclidean Distance $\sqrt{5}$

Manhattan Distance 3

Cosine Distance

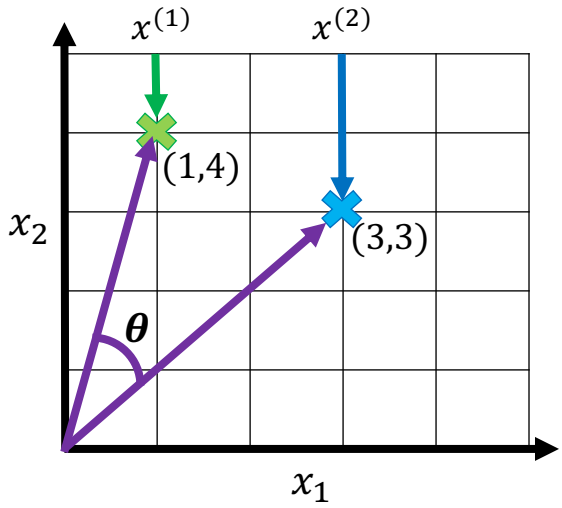
Measures dissimilarity based on the angle between vectors, independent of magnitude

$$1 - \cos\boldsymbol{\theta} \quad \in [\mathbf{0}, \mathbf{2}]$$

Cosine similarity

$$\cos\boldsymbol{\theta} = \frac{x^{(1)} \cdot x^{(2)}}{\|x^{(1)}\| \|x^{(2)}\|} = \frac{\sum_{j=1}^{D} (x_j^{(1)} x_j^{(2)})}{\sqrt{\sum_{j=1}^{D} (x_j^{(1)})^2} \sqrt{\sum_{j=1}^{D} (x_j^{(2)})^2}}$$

Dot product of the two vectors normalized by their magnitudes



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Distance Metrics

Euclidean Distance $\sqrt{5}$

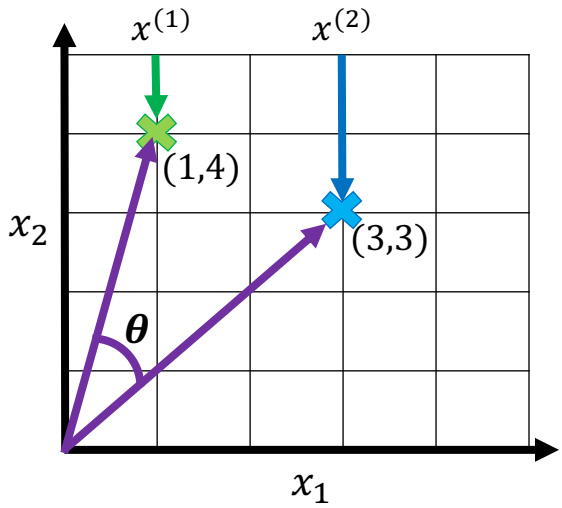
Manhattan Distance 3

Cosine Distance

$$1 - \frac{(1 * 3) + (4 * 3)}{\sqrt{(1)^2 + (4)^2} \sqrt{(3)^2 + (3)^2}}$$

$$1 - 0.857$$

$$0.143$$



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Distance Metrics

Euclidean Distance $\sqrt{5}$

Manhattan Distance 3

Cosine Distance

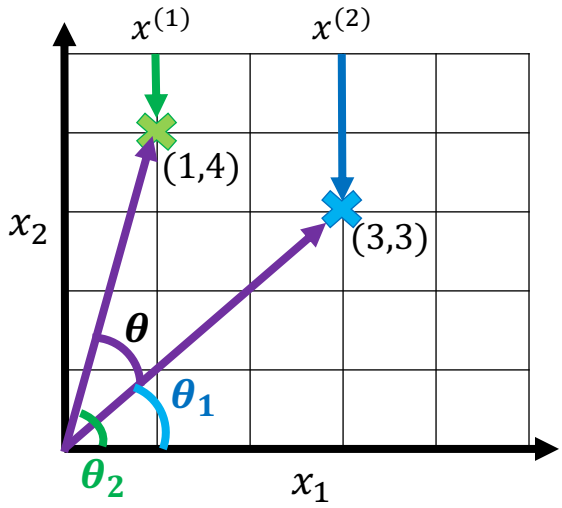
$1 - 0.857 = 0.143$

$\cos(\boldsymbol{\theta_2} - \boldsymbol{\theta_1})$

$\cos(\tan^{-1}(\frac{\mathbf{4}}{\mathbf{1}}) - \mathbf{45})$

$\cos(\mathbf{75.964} - \mathbf{45})$

$\cos(30.964) \approx 0.\mathbf{857}$



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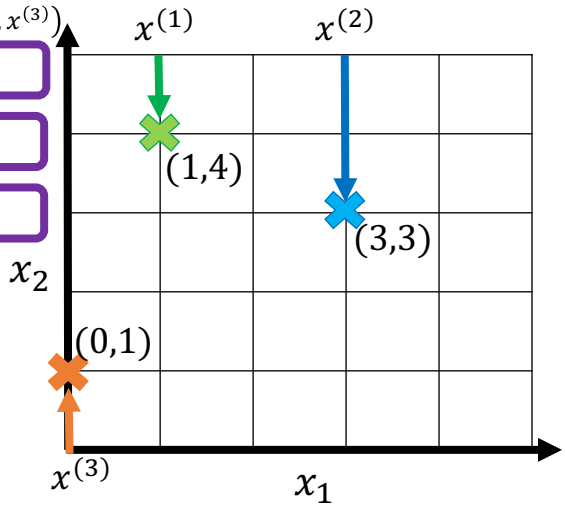
Distance Metrics

	$d(x^{(1)}, x^{(2)})$	$d(x^{(1)}, x^{(3)})$
Euclidean Distance	$\sqrt{5}$	?
Manhattan Distance	3	?
Cosine Distance	0.143	?

Q: Can we say $d(x^{(1)}, x^{(2)}) < d(x^{(1)}, x^{(3)})$?



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Distance Metrics

Q: how to avoid it?

Euclidean Distance
$\| x^{(1)} - x^{(2)} \|_2$

- Most intuitive geometric distance
- Sensitive to magnitude and scale.
- *Common in: K-means clustering, K-NN, physical space measurements*

Manhattan Distance
$\| x^{(1)} - x^{(2)} \|_1$

- Also known as "city block" distance
- Represents grid-based movement
- Less sensitive to outliers than Euclidean distance.
- *Common in: Grid-based problems, city-block navigation, robust clustering*.

Cosine Distance
$1 - \cos \boldsymbol{\theta}$

- Focus on direction/orientation
- Scale-invariant (magnitude doesn't matter)
- *Common in: Text similarity, document retrieval, recommendation systems*

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Distance Metrics for Continuous/numerical Data

Euclidean Distance
$\| x^{(1)} - x^{(2)} \|_2$

Manhattan Distance
$\| x^{(1)} - x^{(2)} \|_1$

Cosine Distance
$1 - \cos \boldsymbol{\theta}$

Q: But what about categorical data/features?
E.g., color, animal type, city, music genre, etc.

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Distance Metrics for Categorical Data

Hamming Distance
Counts mismatches between categorical vectors

$$d_{Hamming}\left(x^{(1)}, x^{(2)}\right) = 4$$

Two users' answers in a 6-question multi-choice quiz:

- User 1 ($x^{(1)}$): [A, B, C, D, A, C]
- User 2 ($x^{(2)}$): [A, C, C, A, B, D]

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Distance Metrics for Categorical Data

Hamming Distance
Counts mismatches between categorical vectors

$$\mathrm{d}_{\mathrm{Hamming}}\left(x^{(1)}, x^{(2)}\right) = 4$$

Jaccard Distance
Based on sets overlap

$$d_{\mathrm{Jaccard}}\left(x^{(1)}, x^{(2)}\right) = 1 - \frac{X^{(1)} \cap X^{(2)}}{X^{(1)} \cup X^{(2)}}$$

Two users' answers in a 6-question multi-choice quiz:

- User 1 ($x^{(1)}$): [A, B, C, B, A, C]
- User 2 ($x^{(2)}$): [A, C, C, A, B, D]

Not well-suited for the above features.

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Distance Metrics for Categorical Data

Hamming Distance
Counts mismatches between categorical vectors

$$\mathrm{d}_{\mathrm{Hamming}}\left(x^{(1)}, x^{(2)}\right) = 4$$

Jaccard Distance
Based on sets overlap

$$d_{\mathrm{Jaccard}}\left(x^{(1)}, x^{(2)}\right) = 1 - \frac{X^{(1)} \cap X^{(2)}}{X^{(1)} \cup X^{(2)}} = 1 - \frac{2}{7} = 0.714$$

Two users' answers in a 6-question multi-choice quiz:

- User 1 ($x^{(1)}$): [A, B, C, B, A, C]
- User 2 ($x^{(2)}$): [A, C, C, A, B, D]

Two users' favorite film types:

- Alice ($x^{(1)}$): {Action, Comedy, Sci-Fi, Thriller}
- Mia ($x^{(2)}$): {Comedy, Drama, Romance, Sci-Fi, Animation}

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Distance Metrics for Categorical Data

Hamming Distance
Counts mismatches between categorical vectors

- Data is ordered and vectors are of the same length
- You care about position-by-position mismatches
- E.g., DNA sequence, multi-choice answers, etc.

Jaccard Distance
Based on set overlap

- Data is set-like (unordered categories, tags, items)
- You care about overlap vs total unique items
- E.g., customer segmentation, shopping baskets, keywords/tags in documents, etc.

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Q1: In the favorite film genre scenario, could we also use Hamming distance? If so, how?

A: Yes. Convert each sample into a fixed binary vector.

$$d_{\text{Hamming}}(\text{Alice}, \text{Mia}) = 5$$

Q2: How can we measure distance when a dataset contains both numerical and categorical features?

A: Gower distance (Weighted distance approach)

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Parametric Models

Linear Regression

Logistic Regression

Assumes a fixed function form with a finite set of parameters
Key idea: Learn parameters θ from data, then discard training data

Non-Parametric Models

K-Nearest Neighbours

Makes minimal assumption about functional form; model complexity grow with data
Key idea: Require training data for predictions, no need to learn parameters.

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Parametric Models

- Linear Regression
- Logistic Regression
- Neural Networks
- Naïve Bayes

Assumes a fixed function form with a finite set of parameters
Key idea: Learn parameters θ from data, then discard training data

Non-Parametric Models

- K-Nearest Neighbours
- Decision Trees
- Random Forests
- Kernel SVM

Makes minimal assumption about functional form; model complexity grow with data
Key idea: Require training data for predictions, no need to learn parameters.

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Bias-variance tradeoff

Models with high bias are prone to **underfitting**, missing important patterns.

Models with high variance are prone to **overfitting**, capturing noises as if it were signal.

- **Bias** - measures how far off, on average, a model's predictions are from the ground truth values

- **Variance** measures how much a model's predictions change with different training datasets.

Striking the right balance is at the heart of effective machine learning design

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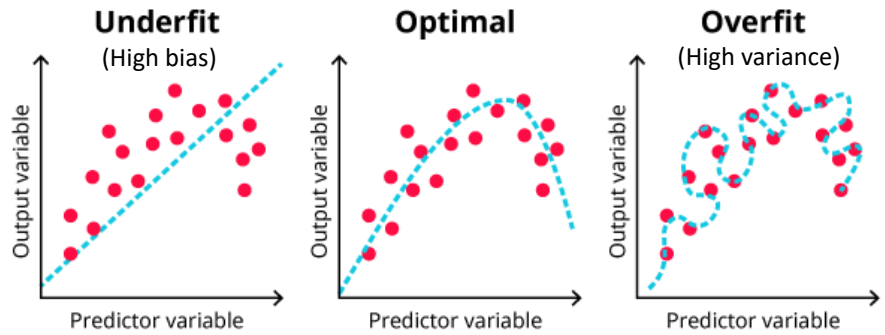
Underfitting and Overfitting

- Too simple
- Does not fully learn the data

- Too complex
- Fits the noise in the training data
- Does not generalize well



Striking the right balance is at the heart of effective machine learning design

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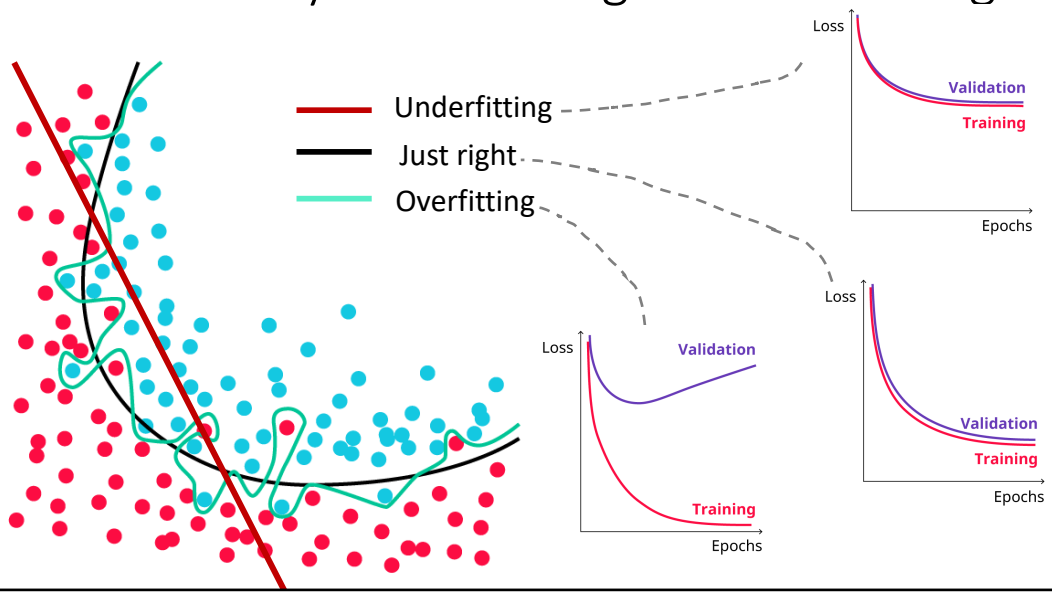
How to Identify Underfitting and Overfitting



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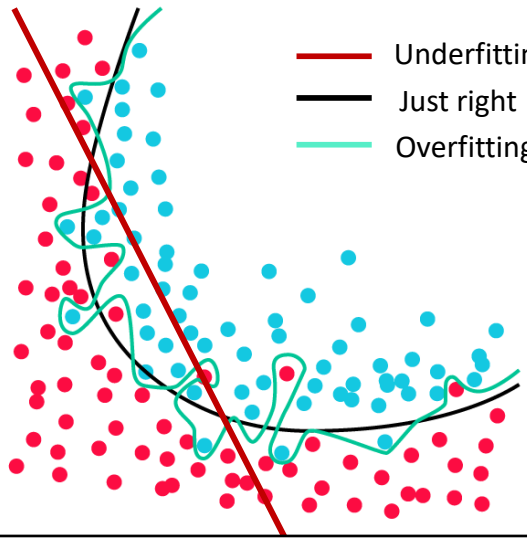
How to Mitigate Underfitting and Overfitting



- More complex model
- Add more relevant features
- Reduce regularization strength

- Use simpler model
- Increase training data
- Apply regularization

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Bias and variance in KNN?

K too small → Sensitive to noise → High variance

K too big → Underfitting (the majority class) → High bias

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Summary

Explain how the k-NN operates for regression and classification tasks.

Describe the advantages and disadvantages of k-NN.

Calculate continuous and categorical distances between two vectors.

Distinguish parametric vs non-parametric, overfitting vs underfitting

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Feedback!

If you have any feedback, use the QR code or link to anonymously report it.

I'm open to any comments/feedback.

SCAN ME

https://forms.gle/AyTC145yPeFVMiQw8

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